

The Impact of Digital Payments and Monetary Policy on Currency Demand in the US: A Johansen Co-integration Analysis

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ABSTRACT

This is an analysis of the dynamics of credit card usage, inflation, and the federal funds rate in the United States and their effect on currency in circulation, using monthly data from 2001 to 2024. Using a descriptive and analytical research design, data were taken from the Federal Reserve Economic Data (FRED). The analysis used Stata Statistical Software for stringent econometric testing.

Stationarity was tested with Augmented Dickey-Fuller (ADF) and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests which showed that variables are integrated of order one, $I(1)$. The Johansen Co-integration Test proved that there is a unique long-run equilibrium relationship among the variables. Estimates, based on the Vector Error Correction Model (VECM), show a significant negative long run elasticity of credit card usage and currency demand, which is strong empirical evidence for the Substitution Hypothesis. Furthermore, Wald Test shows that there is a strong causality from credit cards to currency in the short run ($p = 0.0003$). Structural stability of the model by the CUSUM test was verified for the policy inference of the results.

Keywords: Unit Root Test, VECM, Co-integration, Granger Causality, Currency Demand, Digital Payments, Substitution Hypothesis, ARIMA, Forecasting.

Chapter 1: Introduction

Despite the explosion in digital financial technologies, the demand for physical currency is a hardy and important topic of monetary policy. The United States economy has experienced a structural change in the way people pay, both in terms of the ubiquity of credit card usage and electronic transfers. According to the Federal Reserve's Diary of Consumer Payment Choice, the percentage of cash transactions has fallen off a cliff, but the total value of currency in circulation is still rising.

This development makes us think about the stability of money demand functions and the future of cash as a way to store value and trade. Classical economic theory (more precisely, the Baumol- Tobin model) suggests that the lower the transaction costs of the alternatives to physical currency, the lower the demand for physical currency should be. However, the continued growth of currency in the face of digital innovation is a notable empirical puzzle that has been termed the “Cash Paradox”.

The macroeconomic picture is further compounded by periods of considerable volatility such as the 2008 Financial Crisis and the epidemic of Covid-19, which delivered enormous external shocks to the monetary aggregates. During these crises, cash tends to return to its role as a safe store of value, and so the transactional trend of digital payments. Consequently, the underlying aim of this research is to address the following question: What is the dynamic relationship between Currency in Circulation, Credit Card Usage, CPI (Inflation) and Federal Funds Rate? The main goal is to measure the role of digital payments and monetary policy variables in explaining currency demand in the US. The methodology involves Johansen Co-integration Test after standard unit root testing, Vector Error Correction Model (VECM), Wald Causality Tests to separate the long run relationships from the short run dynamics and ARIMA for forecasting Currency in Circulation for next 3 years.

Chapter 2: Literature Review

Recent scientific study and a number of economic ideas make an effort to comprehend how money demand is evolving. This study, which incorporates research from both local and global

settings, focuses on the relationship between digital innovation, financial literacy, crisis-related decisions, and institutional frameworks.

2.1. The Velocity of Money and Digital Innovation

Classical theory has it that because payment innovations remove the need for money balances, the velocity of money should rise. Liu, Li, and Wang (2021) empirically tested this in the Chinese market and found that digital payments positively and significantly affect the velocity of money (M2 level). Their study refutes traditional models that do not consider fintech variables by maintaining that digital platforms increase the frequency of transactions. This is further supported by Riedler (2017), who showed the extent to which electronic systems such as M-Pesa and Bitcoin reduce transaction costs (1.4% fees in comparison to traditional channels) thereby affecting international money transfer behavior and eliminating the friction that generally necessitates cash holdings. Similarly, Abednego and Apriansah (2010) observed in Indonesia that the money demand function is changed by electronic

money as a result of reduction in average cash holdings and so increases the velocity of circulation in the economy.

2.2. The Substitution Hypothesis vs. The Cash Paradox

The "Substitution Hypothesis" holds that digital payments replace cash. Amromin and Chakravorti (2009) were among the first to isolate small-denomination notes finding debit cards significantly destroyed demand for transactional cash. Building upon this, Schoellner (2002) found that the more credit card is used, the more cash will be reduced and more credit card balance in checking account.

However, a "Currency Demand Paradox" has arisen in the world. Dhal et al. (2023) studied this in India, and there was a simultaneous increase in digital payments and currency in circulation (CiC). They found that although digital payments do have a negative substitution effect on cash, the size of the effect is smaller than the positive income effect from GDP growth. This is similar to the situation in the U.S. by Coyle et al. (2022), who found that during 2017-2020, although the use of cash has been growing less rapidly than the use of checks, older and lower income demographics still relied heavily on physical currency to sustain aggregate demand. Slozko and Pelo (2014) brings a developmental viewpoint into the picture arguing that the transition to

cashless payments is strongly related to economic infrastructure; therefore, the effect of substitution is not even across different levels of GDP per capita.

2.3. Crisis, Culture, and Uncertainty Avoidance

The survival of cash is often precautionary, especially in times of crisis. Kotkowski (2022) examined 58 countries around the world throughout the pandemic and found "uncertainty avoidance" to be the best predictor of cash hoarding. High uncertainty cultures hoarded more cash even though it was available digitally. This is consistent with the finding of Davoodalhosseini (2024), who found that the demand for cash in the pandemic was precautionary in nature rather than transactional and driven by the fear of contagion and economic anxiety.

Behavioural response towards unpredictability are also impacted by literacy. Sadok and Elouaourti took a very interesting look at the "cash vs. digital paradox" with the 2022 PISA data. They learned that financial literacy works as a buffer against cash reversion; digitally literate people continued to use digital payment habits even during uncertain times while those that were less literate reverted to cash.

2.4. Financial Literacy and Adoption Barriers

The impact of digital substitutes is very much dependent on human capital. Dahlberg et al. (2022) found in Norway that there is a correlation between higher financial literacy and a lower level of financial vulnerability among mobile payment users. This flies in the face of the idea that digital payments actually encourage people to spend more by removing the "pain of paying," i.e., that the more financially sophisticated one is, the less of a "pain" paying feels. On the other hand, in the developing markets, there are still high barriers. Hassan et al. (2023) revealed that in Pakistan, trust and security issues are major inhibitors against the Digital Financial Services (DFS) adoption, where cost and poor infrastructure are the inhibiting factors for the users in rural areas.

2.5. Institutional and Legal Perspectives

Finally, the demand for currency is an institutional trust function. The Federal Reserve Bank of New York (2021) highlights that money is a "legal construct" which is founded on trust. While there is efficiency to be found in new tools such as stablecoins, we run the risk of re-creating the

instability of the "Free Banking" era with none of the safety benefits of the tiered Fed-bank system. This legal view is in support of the ESTA (2018) point that cash is an important "public good." They suggest, thus, that cash is crucial for contingency for digital failures and cyber threats, so the demand for physical currency is likely to remain as a safety net even as transactional demand disappears. Rogoff (2016) provides a counter-narrative in "The Curse of Cash," and argues that the lasting demand for large-denomination notes (\$100 bills) is driven more by the underground economy than by real commerce, which would seem to suggest that the "public good" argument does have its limits.

Chapter 3: Material and Methods Used

3.1. Research Design

The study uses a quantitative, descriptive, and analytical approach to research. Secondary time-series data was used to measure the selective impact of the independent variables on the dependent variable. Every econometric experiment was done using Stata software for statistics.

3.2. Conceptual Framework and Variables

The study adopts a standard money demand function adjusted with a alternative for payment technology. The functional is goig to be expressed as:

$$\ln(\text{CURRCIR})_t = \beta_0 + \beta_1 \ln(\text{CreditCards})_t + \beta_2 \ln(\text{CPI})_t + \beta_3 \text{FEDFUNDS}_t + \epsilon_t$$

Table 3.1: Variable Description and Measurement

Variable	Definition	Measurement
ln_CURRCIR	Currency in Circulation	Natural Log of USD Trillions
ln_CreditCards	Commercial Bank Credit Card Plans	Natural Log of Volume
ln_CPI	Consumer Price Index	Natural Log of Index
FEDFUNDS	Effective Federal Funds Rate	Percent

3.3. Nature and Sources of Data

The data used in this analysis are secondary time series. The primary source is the Federal Reserve Economic Data (FRED) database, which is supported by the official releases from the Bureau of Labor Statistics data that was procured.

3.4. Time Period

To cover a wide range of economic conditions, we analyse around 286 sets of monthly time series data covering the whole period from 2001 till 2024.

3.5. Econometric Method

The following sequence of procedures was used to ensure accurate and reliable estimation:

3.5.1. Stationarity Test

Standard a regression study needs data to stay stable to avoid "spurious regression." A stationary series contains a fixed mean and variation throughout time. In order to get stationarity, variables integrated of order one, $I(1)$, that have unit roots, need to be defined in first differences

$$(\Delta y_t = y_t - y_{t-1}).$$

Tests Employed: Augmented Dickey-Fuller (ADF) Test and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) Test.

3.5.2. Co-integration Test (Johansen)

If a linear combination of non-stationary variables is stationary, the series are said to be co-integrated, with the implication of a valid long-run relationship, in the sense of an equilibrium relationship. The Johansen Co-integration test is used to find out the number of such relationships in a multivariate setting. This approach is superior to the Engle-Granger two-step approach since it is able to identify multiple cointegrating vectors and more robust to endogeneity.

3.5.3. Vector Error Correction Model (VECM)

While co-integration shows long-run relationships, the VECM shows dynamic adjustments. It captures the speed of the system in the recovery of disequilibria and return to the long run path, and also estimates the short run causalities. This model is especially useful for analyzing the impulse response of currency demand to shocks in the use of credit cards.

3.5.4. Causality and Impulse Response Analysis

To find out the direction and time of relationship between variables, this study uses Granger Causality tests (Wald tests). While co-integration establishes a long-run connection, the tests of causality are important in determining whether past values of the use of credit card provide statistically significant information for the prediction of future demand for currency in the short-run. In addition, Impulse Response Functions (IRFs) are also used for recording the system's changing course. IRFs depict the reaction of currency demand to a certain one-time shock in credit card usage, including the magnitude as well as the persistence of the shock over time.

3.5.5. Univariate Forecasting (ARIMA)

To depict the future trend of the dependent variable, a univariate model of Auto-Regressive Integrated Moving Average (ARIMA) is constructed according to the Box-Jenkins approach. Unlike the structural VECM, this approach is concerned only with the probabilistic qualities of the currency time series itself. By iteratively finding the optimal autoregressive and moving average terms, the ARIMA model captures the historical inertia of the data to produce robust forecasts out-of-sample of data up to 2027 to forecast currency demand.

3.6. Diagnostics

The following Diagnostics were used for the diagnostics of dataset.

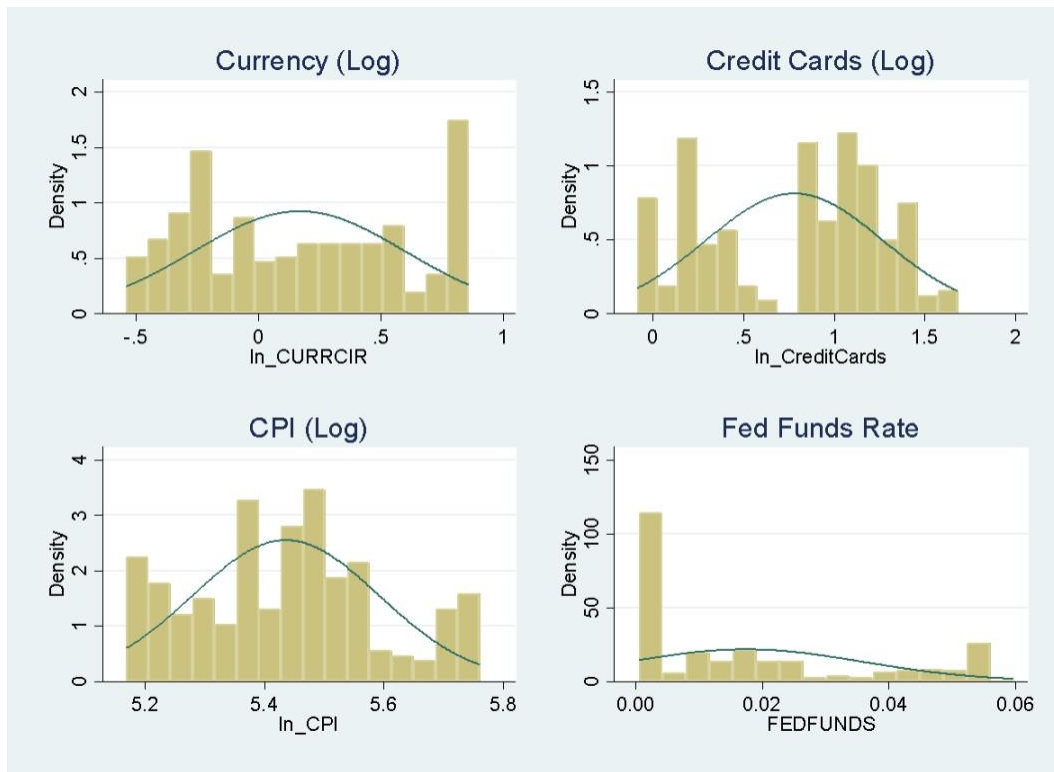
3.6.1. Descriptive Statistics

Table 3.1 shows the descriptive statistics of the logged variables. The low values of skewness, especially for $\ln_CURRCIR$ (0.164), suggest that the log transformation was successful in stabilising the distributions.

Table 3.1: Descriptive Statistics

Variable	Mean	Median (p50)	Std. Dev.	Min	Max	Skewness	Kurtosis
ln_CURRCIR	0.168	0.147	0.433	-0.540	0.859	0.164	1.716
ln_CreditCards	0.780	0.896	0.492	-0.088	1.682	-0.279	1.728
ln_CPI	5.436	5.445	0.157	5.168	5.761	0.202	2.365
FEDFUNDS	0.017	0.011	0.019	0.001	0.060	0.876	2.322

Graph 3.1



3.6.2. Correlation and Multicollinearity Analysis

Before moving on to the deeper analysis of this data, we verified the relationships between the level variables. Credit cards and CPI share a strong relationship (0.93), according to the pairwise correlation matrix (Table 3.2).

Table 3.2: Pairwise Correlation Matrix

	ln_CURRCIR	ln_CreditCards	ln_CPI	FEDFUNDS
ln_CURRCIR	1.0000			
ln_CreditCards	0.9312*	1.0000		
ln_CPI	0.9751*	0.9277*	1.0000	
FEDFUNDS	-0.0474	-0.1401*	0.0173	1.0000

3.6.3. Multicollinearity Check

Variance Inflation Factors (VIF) have been calculated with the intention to measure the danger that this relationship has to model stability. The strong collinearity among credit cards and CPI can be seen in Table 3.3, making the use of initial differences in further estimations vital

Table 3.3: Variance Inflation Factor (VIF)

Variable	VIF	1/VIF
ln_CreditCards	9.26	0.1079
ln_CPI	9.10	0.1099
FEDFUNDS	1.29	0.7738
Mean VIF	6.55	

3.6.4. Diagnostic Tests (Levels Estimation)

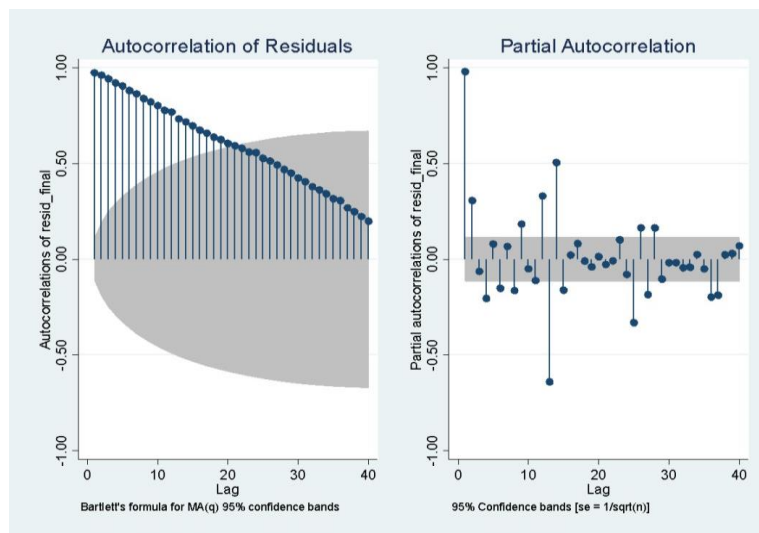
Initial OLS estimation on the level variables highlighted several specification issues common in time- series data. As summarized in Table 3.4, the model failed tests for homoscedasticity and

serial correlation, confirming that a simple static model is insufficient and a dynamic approach (VECM) is required.

Table 3.4: Diagnostic Test Summary (on Levels)

Test	Null Hypothesis (H0)	Test Statistic (Chi2/W)	Prob.	Conclusion
White's Test	Homoscedasticity	$Chi^2 = 56.04$	0.0000	Reject H_0 (Heteroscedasticity)
Breusch-Godfrey	No Serial Correlation	$Chi^2 = 276.63$	0.0000	Reject H_0 (Autocorrelation)
Shapiro-Wilk	Normal Distribution	$W = 0.967$	0.0000	Reject H_0 (Approx. Normal)

Graph 3.2: AR/PA



3.6.5. Unit Root Tests

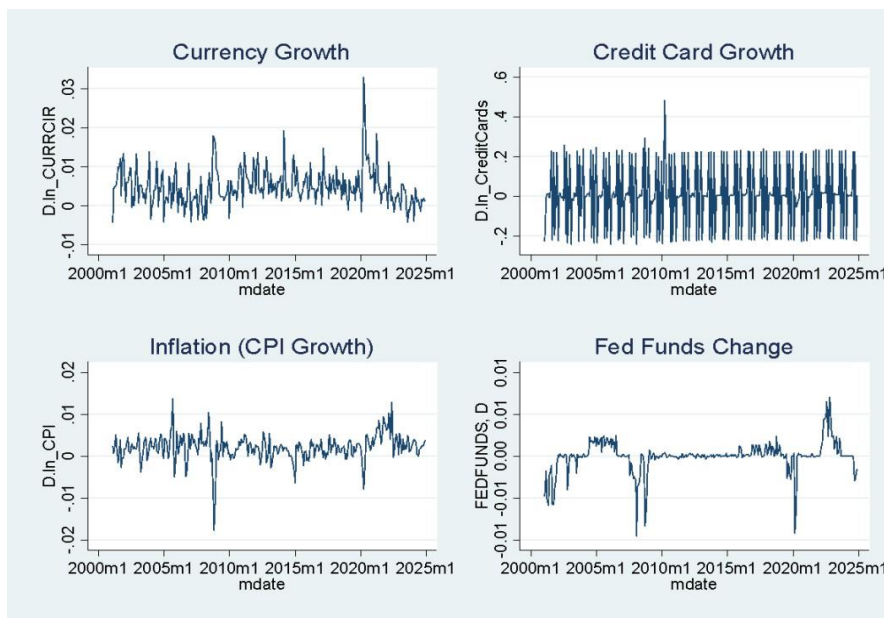
Variance Inflation Factors (VIF) have been calculated with the intention to measure the danger that this relationship has to model stability. The strong collinearity among credit cards and CPI can be seen in Table 3.5, making the use of initial differences in further estimations vital.

Table 3.5: Result of Unit Root Tests

Variable	ADF (Levels)	KPSS (Levels)	ADF (1st Diff)	Conclusion
ln_CURRCIR	Fail to Reject	Reject ***	Reject *** ($p = 0.000$)	$I(1)$
ln_CreditCards	Reject ***	Reject ***	Reject *** ($p = 0.000$)	$I(1)$
ln_CPI	Fail to Reject	Reject ***	Reject *** ($p = 0.000$)	$I(1)$
FEDFUNDS	Fail to Reject	Reject ***	Reject *** ($p = 0.000$)	$I(1)$

All variables are Integrated of Order One, $I(1)$. This satisfies the requirements to apply Johansen Co-integration Test.

Graph 3.3: Stationarity



3.6.6. Post-Differencing Multicollinearity Check:

After changing the variables to first differences (growth rates) we remeasured the Variance Inflation Factors (VIF) to make sure the previously identified multicollinearity was solved. The high VIF for the level variables (9.26) was caused by the commonality in the long-term trend between prices and credit usage. By differencing the data we effectively took away this common trend and isolated the independent variation of each series. Consequently, the VIF fell drastically to 1.11, which is well below the threshold of 5. This ensures that multicollinearity is no longer a problem and that the coefficient estimates in our dynamic model are stable and reliable.

Table 3.6: Result of VIF after Transformation (1st difference)

Variable	VIF	1/VIF
D1. FEDFUNDS	1.09	0.914387
D1. ln_CPI	1.09	0.918254
D1. ln_CreditC~s	1.01	0.994480
Mean VIF	1.06	

Chapter 4: Analysis and Interpretation

4.1. Co-integration Analysis

The Johansen methodology was used to determine the number of co-integrating vectors (r) using the Trace statistic.

Table 4.1: Result of the Johansen Co-integration Test (Trace)

Hypothesized CE(s)	No. of	Eigenvalue e	Trace Statistic	0.05 Critical Value	Prob.	Result
None ($r = 0$)		-	65.59	47.21	0.000	Reject
At most 1 ($r = 1$)		0.1198	29.10	29.68	> 0.05	Fail to Reject

At most 2 ($r = 2$)	0.0758	6.56	15.41	> 0.05	Fail to Reject
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Trace test indicates 1 co-integrating eqn(s) at the 0.05 level.

The Trace statistic for the null hypothesis of no co-integration ($r = 0$) is 65.59, exceeding the critical value of 47.21. So, we may reject the null hypothesis and conclude that one unique cointegrating equation exists.

4.2. Vector Error Correction Model (VECM)

Given the long-run relation, a VECM was estimated. The cointegration equation showed a highly negative long-run relationship.

Cointegrating Equations (_ce1)

Summary Statistics:

- Chi2: 41.91479
- Prob > Chi2: 0.0000

Table 4.2: Result of the Vector Error Correction Model (VECM)

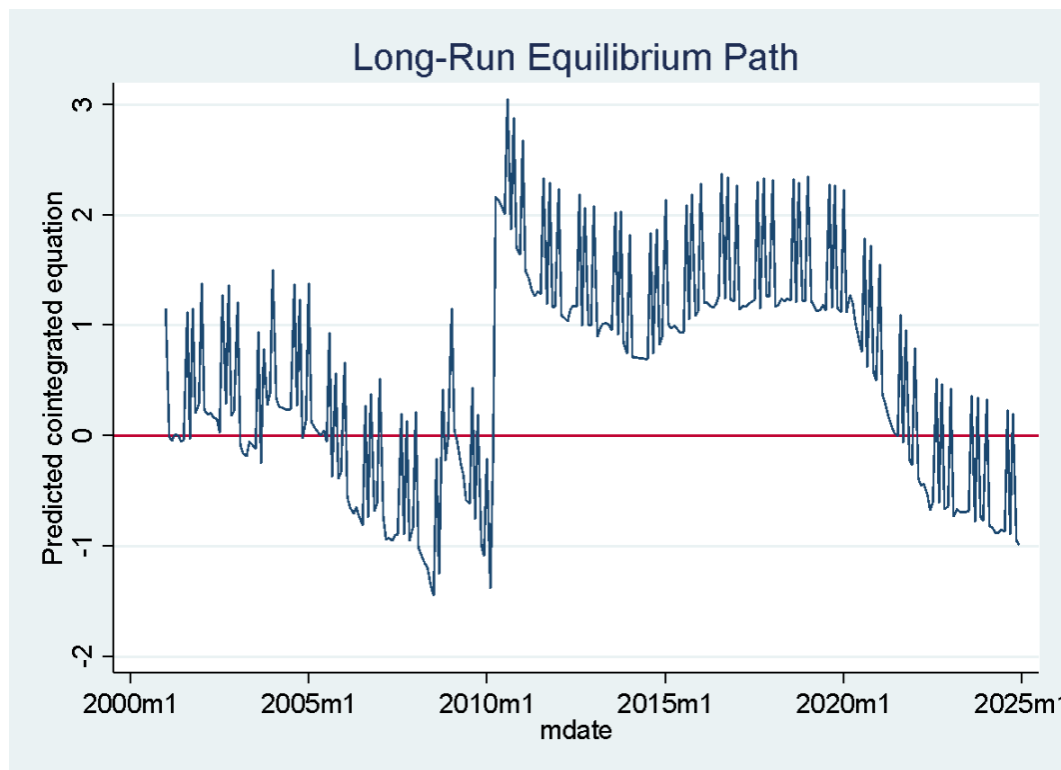
Beta	Coefficient	Std. Err.	z	P> z	[95% Conf. Interval]
ln_CURRCIR	1	.	.	.	.
ln_CreditCards	4.972568	1.019178	4.88	0.000	2.975015 — 6.97012
ln_CPI	-17.08382	3.164693	-5.40	0.000	-23.28651 — -10.88114
FEDFUNDS	-3.221095	9.63688	-0.33	0.738	-22.10903 — 15.66684
_cons	89.44094	.	.	.	.

The normalized long run coefficient for $\ln_CreditCards$ was estimated to be -4.97 ($p < 0.01$). This indicates that a 1% rise in the use of credit cards is linked to a fall (in the long run) of around 5% of the demand for currency. This finding provides empirical validity to the Substitution Hypothesis, that digital payments substitute for physical cash given time.

To interpret the economic relationship, we move the independent variables to the right-hand side of the equation, which reverses their signs:

$$\ln(CURRCIR) = -4.97 \ln(CreditCards) + 17.08 \ln(CPI) + 3.22(FEDFUNDS) - 89.44$$

Graph 4.1: Long Run graph



Graph 4.1 illustrates the long-run equilibrium deviations with a strong focus on significant structural imbalances that have occurred during the 2009 Global Financial Crisis and the COVID-19 pandemic of 2020. In spite of these long periods of disequilibrium, the series has mean-reverting behavior, and is gradually moving back to the zero level. Thus, the data is

consistent with the opinion that despite the fact that exogenous shocks have a short-term effect of disequilibrium in the system, there are strong forces in the economy that eventually restore the long-run correlation between credit usage, currency, and prices.

Short term analysis

In short run, the deviations from equilibrium are temporary. The system has an error correction mechanism that pulls the variables back to long run path.

4.3. Short Run Causality (Wald Test)

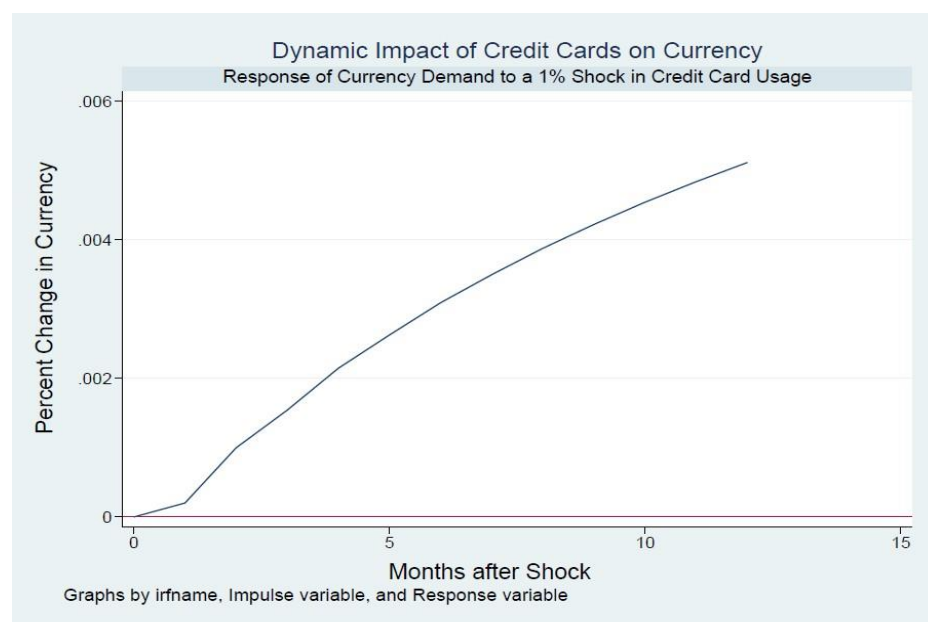
To verify the direction of causality within the co-integrated system, the Wald Test was applied to the short-run coefficients (lagged differences).

Table 4.3: Wald Test Results

Dependent Variable	Causal Variable	Chi-Square	Prob.	Conclusion
$\Delta \ln(\text{Currency})$	$\Delta \ln(\text{CreditCards})$	12.96	0.0003	Causality Exists

Table 4.3 shows us that the p-value is 0.0003, thus, there is strong evidence of short-run causality running from Credit Cards to Currency. This shows that an increase in credit card usage acts as a leading factor for a decline in currency circulation.

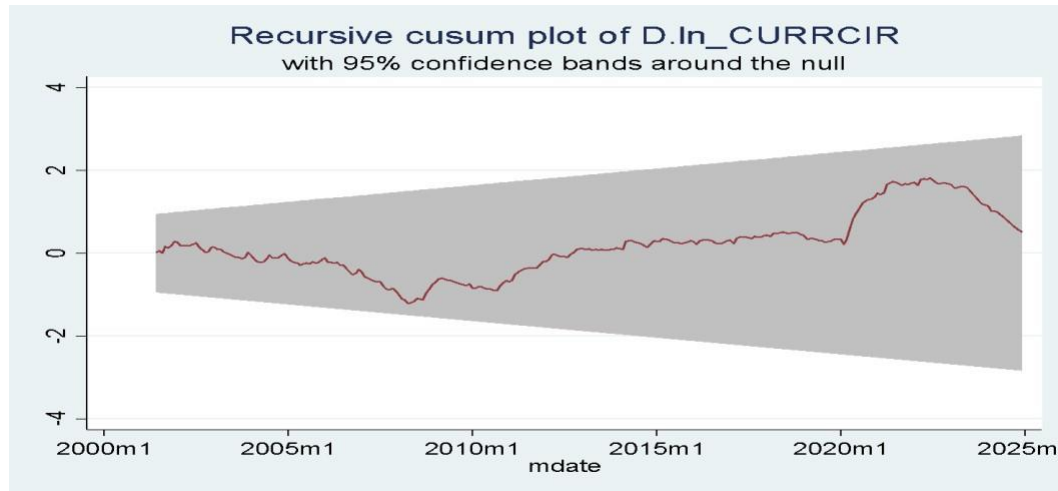
Graph 4.2: IRF



4.4. Structural Stability

To verify the stability of the long-run coefficients, the Cumulative Sum of Recursive Residuals (CUSUM) test was applied.

Graph 4.3



Graph 4.3 shows that the test statistic does not exceed the 95% critical values during the study period. This proves that the model is structurally stable regardless of major economic events, including the 2008 Financial Crisis and the pandemic in 2020 (Covid-19).

4.4 Univariate Forecasting with ARIMA

4.1 Lag Selection


```
. varsoc ln_CURRCIR, maxlag(24)
```

Lag-order selection criteria

Sample: 2003m1 thru 2024m12

Number of obs = 264

Lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	-135.475				.164643	1.0339	1.03935	1.04745
1	1039.04	2349	1	0.000	.000023	-7.85638	-7.8455	-7.82929
2	1065.99	53.895	1	0.000	.000019	-8.05296	-8.03663	-8.01232
3	1066.07	.15989	1	0.689	.000019	-8.04599	-8.02421	-7.9918
4	1068.65	5.1686	1	0.023	.000019	-8.05799	-8.03077	-7.99026
5	1070.11	2.904	1	0.088	.000018	-8.06141	-8.02875	-7.98014
6	1070.11	.01676	1	0.897	.000019	-8.0539	-8.0158	-7.95908
7	1072.17	4.1048	1	0.043	.000018	-8.06187	-8.01833	-7.95351
8	1072.75	1.157	1	0.282	.000019	-8.05868	-8.00969	-7.93677
9	1073.61	1.728	1	0.189	.000019	-8.05765	-8.00322	-7.9222
10	1074	.7858	1	0.375	.000019	-8.05305	-7.99318	-7.90405
11	1075.39	2.7653	1	0.096	.000019	-8.05595	-7.99063	-7.89341
12	1079.58	8.3837	1	0.004	.000018	-8.08013	-8.00937	-7.90404
13	1098.65	38.152	1	0.000	.000016	-8.21707	-8.14087	-8.02743
14	1123.79	50.269	1	0.000	.000013*	-8.39991*	-8.31826*	-8.19673*
15	1123.81	.04178	1	0.838	.000013	-8.39249	-8.3054	-8.17576
16	1124.05	.47516	1	0.491	.000013	-8.38671	-8.29418	-8.15644
17	1124.14	.19501	1	0.659	.000013	-8.37987	-8.2819	-8.13606
18	1125.34	2.3833	1	0.123	.000013	-8.38133	-8.27791	-8.12397
19	1125.97	1.2701	1	0.260	.000013	-8.37856	-8.2697	-8.10766
20	1126.27	.60749	1	0.436	.000014	-8.37329	-8.25899	-8.08884
21	1128.63	4.7098*	1	0.030	.000013	-8.38355	-8.26381	-8.08556
22	1128.64	.03029	1	0.862	.000013	-8.37609	-8.2509	-8.06455
23	1129.01	.72763	1	0.394	.000014	-8.37127	-8.24064	-8.04619
24	1129.93	1.8371	1	0.175	.000014	-8.37065	-8.23458	-8.03202

* optimal lag

Endogenous: ln_CURRCIR

Exogenous: _cons

All of the criteria (AIC, HQIC, SBIC & FPE) specify that the optimal lag is 14th lag.

4.2. ARIMA MODEL

To project future currency demand, we estimated an **ARIMA(3,1,3)** model using the Box-Jenkins methodology. This specification was selected because it was the only model where all Auto- Regressive and Moving Average coefficients were statistically significant ($p < 0.05$), satisfying the principle of parsimony.

ARIMA regression

Sample: 2001m2 thru 2024m12

Number of obs = 287

Wald chi2(6) = 2144.61

Log likelihood = 1078.325

Prob > chi2 = 0.0000

D.CURRCIR	OPG		z	P> z	[95% conf. interval]	
	Coefficient	std. err.				
CURRCIR						
_cons	.0058835	.0020885	2.82	0.005	.0017901	.0099768
ARMA						
ar						
L1.	-.3453955	.1237861	-2.79	0.005	-.5880118	-.1027793
L2.	.9089357	.0377542	24.08	0.000	.8349389	.9829326
L3.	.2892245	.1047501	2.76	0.006	.083918	.494531
ma						
L1.	.894465	.120009	7.45	0.000	.6592516	1.129678
L2.	-.6126085	.1066441	-5.74	0.000	-.8216271	-.40359
L3.	-.5731451	.0564215	-10.16	0.000	-.6837291	-.462561
/sigma	.0056354	.0001382	40.77	0.000	.0053645	.0059063

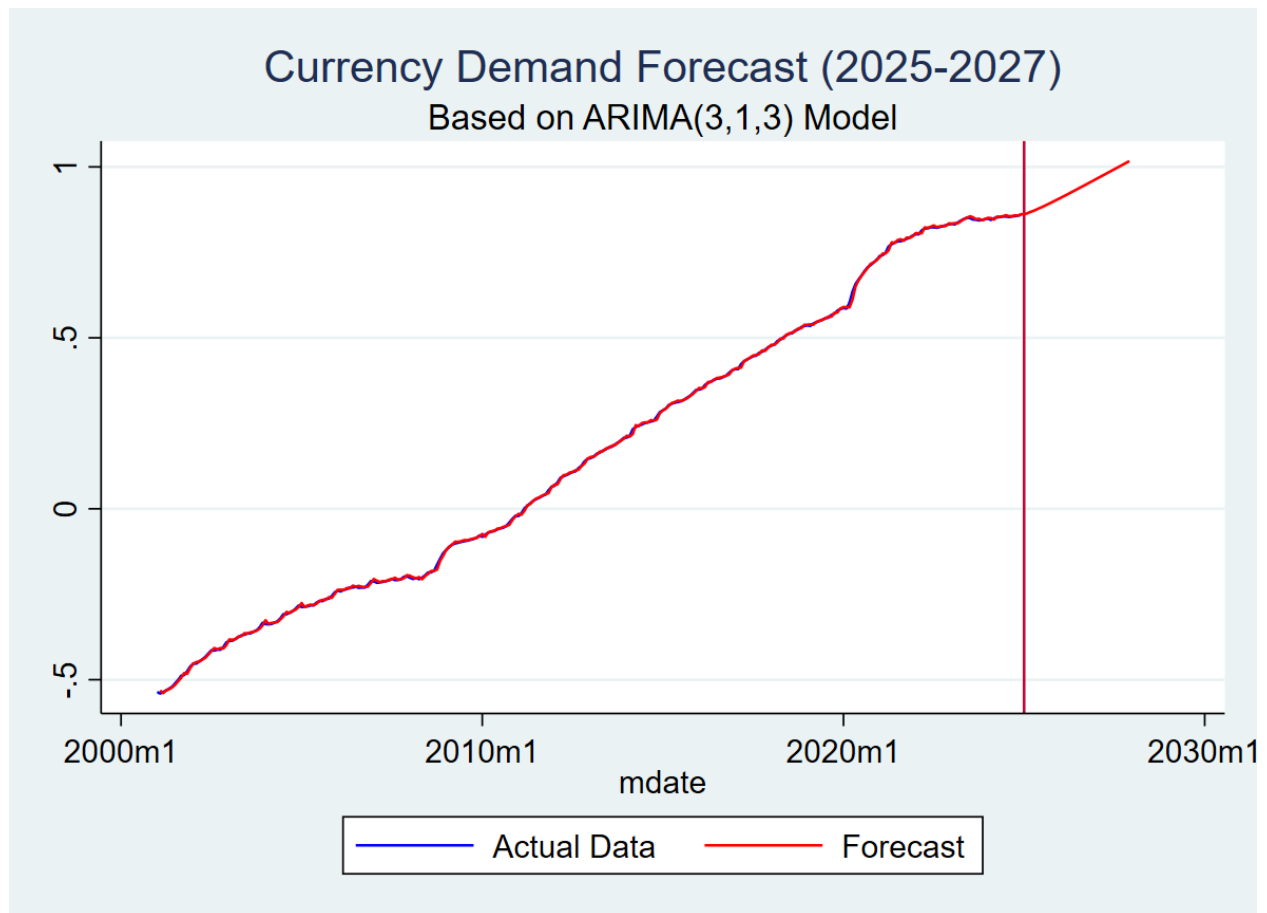
Note: The test of the variance against zero is one sided, and the two-sided confidence interval is truncated at zero.

This forecast does not contradict our earlier finding that credit cards have a negative impact. Instead, it highlights the distinction between marginal effects and aggregate trends:

- **The Brake:** Credit cards exert a significant negative pressure on cash demand (substitution effect).
- **The Momentum:** Inflation and economic growth exert a massive positive pressure (scale effect).

The results suggest that credit cards act as a "brake" slowing the rate of cash growth rather than a "reverse gear." The macroeconomic momentum of the US economy is currently strong enough to overpower the substitution effect of payment technologies in the medium term.

Graph 4.4: Forecasting using ARIMA



ARIMA forecast reveals that the amount of currency in circulation in the U.S. will keep growing between 2025 and 2027 and this is in line with the historical pattern of currency circulating in the country. But the slope of the forecasted values gets observed to be quite flatter than the previous years. This moderation is consistent with the VECM long-run returns, which show that there is a negative correlation between credit card use and cash demand ($b = -4.97$).

Though an increase in credit card payments and this is decreasing the long-run requirements of physical money, more macroeconomic factors such as the growth of the economy, precautionary cash holdings, and international demand of the U.S. dollar are driving aggregate money in circulation up. Therefore, the projection indicates no downward cash flow, but a less steep and more stabilized rate of growth, which will indicate a gradual replacement of cash payments by digital payments in the American economy.

Chapter 5: Conclusion

5.1. Answering the Research Question

The main aim of this paper was to provide the answer to the following question: What is the dynamic relationship between Currency in Circulation, Credit Card Usage, CPI (Inflation), and the Federal Funds Rate?

This question has been answered in the affirmative based on the rigorous econometric analysis that is done. The findings confirm that the correlation between the two variables is not only correlational, but causal and that the structural tension between the payment technology and the macroeconomic scale exists.

- **Long-Run Substitution:** The Vector Error Correction Model (VECM) revealed that there was a statistically significant negative long-run correlation between credit card usage and currency demand. In particular, a 1% rise in credit card volume can be linked to a corresponding 5 percent decline in currency demand otherwise everything else being equal. This is a great empirical support to the Substitution Hypothesis, which states that digital payments provide a substitute to physical cash in the long run.
- **Short-Run Causality:** The Wald Test proved that there is a strong one-way causality between Credit Cards Currency (). This shows that the alterations in payment behavior are a leading indicator to the alterations in the cash demand justifying the predictive ability of the model.
- **Macroeconomic Dominance:** While credit cards exert downward pressure, the analysis revealed that Inflation (CPI) maintains a near-perfect positive correlation () with currency demand. This suggests that the transactional demand for money (driven by price levels) remains a dominant force that counters the substitution effect.

5.2. Reconciling the "Cash Paradox"

A critical contribution of this study is the reconciliation of the negative substitution effect found in the regression analysis with the positive growth trend predicted by the ARIMA forecast.

The findings suggest that the US economy is currently experiencing a "Brake vs. Momentum" dynamic:

- **The Brake (Credit Cards):** The rise of digital payments is actively slowing the velocity of cash circulation, creating a drag on demand.
- **The Momentum (Scale Effects):** However, the sheer scale of economic expansion and inflationary pressure creates a "monetary momentum" that overpowers this brake.

Thus, credit cards are effectively replacing cash in certain types of transactions but they have not yet achieved a level that will result in an irreversible fall in the total stock of currency. The currency does not continue to exist as the favourite medium of exchange, as the nominal worth of the economy grows more quickly than it can be replaced by digital means.

5.3. Policy Implications

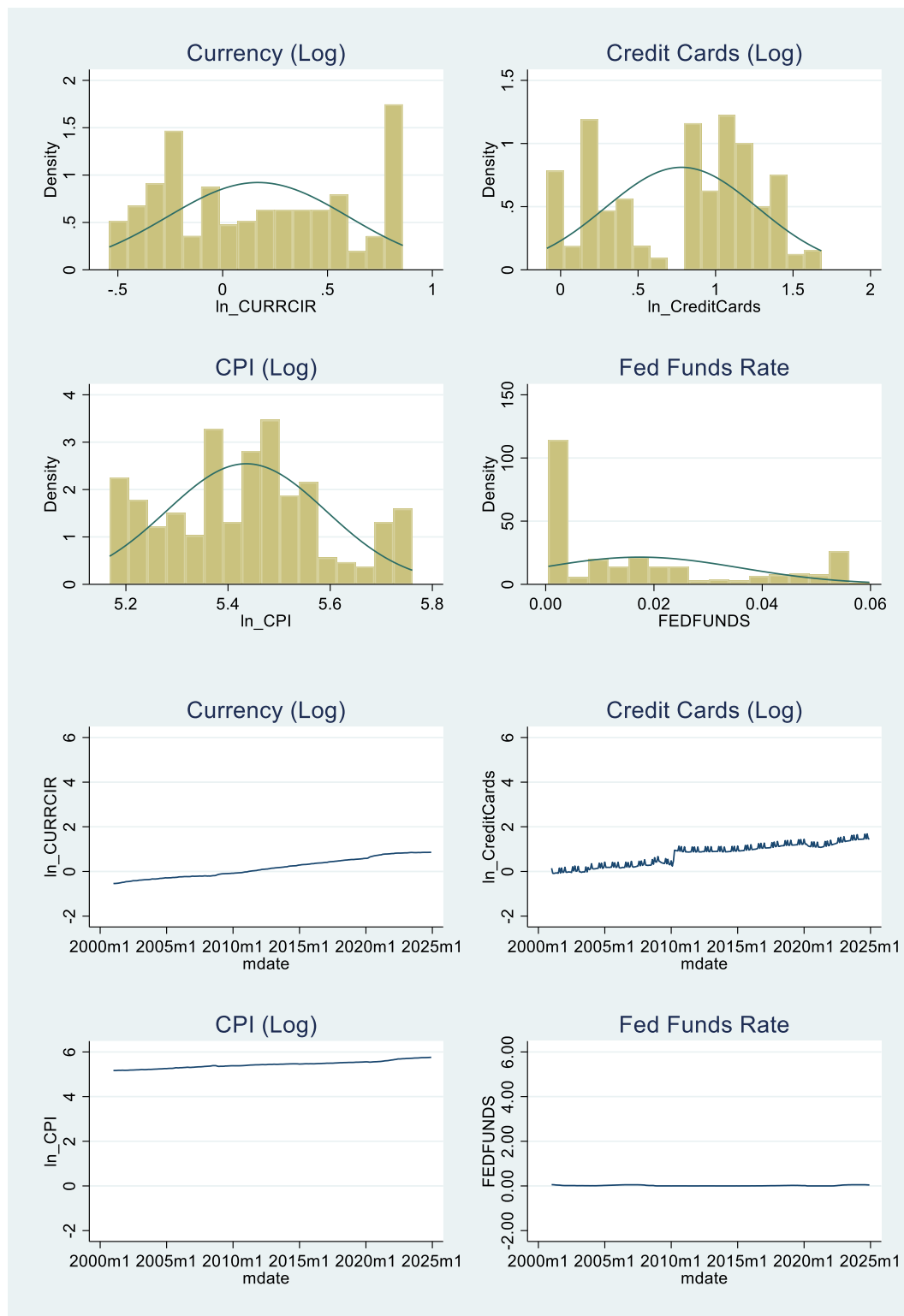
The stability of this relationship as evidenced by the CUSUM test suggests the Federal Reserve cannot assess economic activity using the conventional monetary aggregates only. The central bank digital currency (CBDCs) is a justified concept due to the transition to a less-cash (though not cashless) economy. To bridge this disparity, a digital dollar would be required, with all the efficiency of the credit networks that are now replacing cash and keeping the central bank money public.

To sum up, this study proves that the death of cash is not on hand. Rather, cash has become a strong store of value that expands along with the digital economy, diminished, but not eliminated, by the emergence of the credit card economy.

Appendices and Supplementary Material

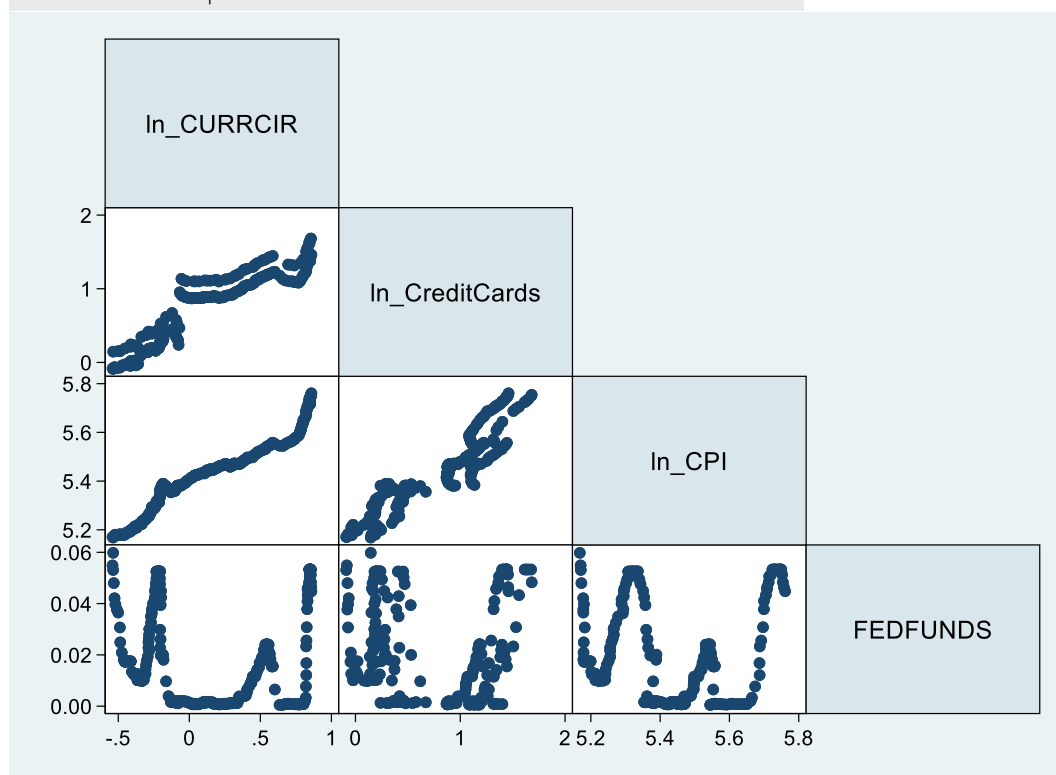
Descriptive Analysis

Variable	Mean	p50	SD	Min	Max	Skewness	Kurtosis
ln_CURRCIR	0.168	0.147	0.433	-0.540	0.859	0.164	1.716
ln_CreditC~s	0.780	0.896	0.492	-0.088	1.682	-0.279	1.728
ln_CPI	5.436	5.445	0.157	5.168	5.761	0.202	2.365
FEDFUNDS	0.017	0.011	0.019	0.001	0.060	0.876	2.322



Correlation

	ln_CUR~R	ln_Cre~s	ln_CPI	FEDFUNDS
ln_CURRCIR	1.0000			
ln_CreditC~s	0.9312* 0.0000	1.0000		
ln_CPI	0.9751* 0.0000	0.9277* 0.0000	1.0000	
FEDFUNDS	-0.0474 0.4231	-0.1401* 0.0173	0.0396 0.5029	1.0000



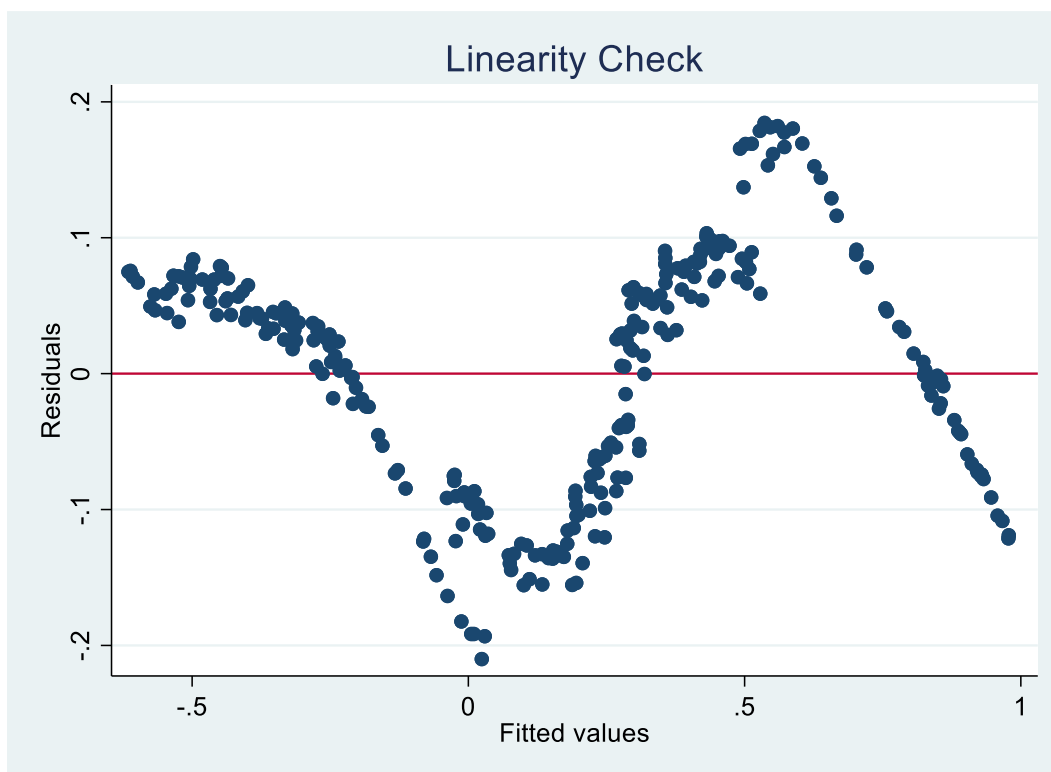
Basic Regression

```
. regress ln_CURRCIR ln_CreditCards ln_CPI FEDFUNDS
```

Source	SS	df	MS	Number of obs	=	288
Model	51.5840959	3	17.1946986	F(3, 284)	=	2236.76
Residual	2.18320239	284	.007687332	Prob > F	=	0.0000
				R-squared	=	0.9594
				Adj R-squared	=	0.9590
Total	53.7672983	287	.187342503	Root MSE	=	.08768

ln_CURRCIR	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
ln_CreditCards	.0927095	.0320433	2.89	0.004	.0296369	.1557821
ln_CPI	2.429981	.0995723	24.40	0.000	2.233987	2.625974
FEDFUNDS	-1.574363	.3170535	-4.97	0.000	-2.198436	-.9502899
_cons	-13.0854	.5152575	-25.40	0.000	-14.0996	-12.07119

Linearity



Heteroscedasticity

White's test

H0: Homoskedasticity

Ha: Unrestricted heteroskedasticity

chi2(9) = 56.04

Prob > chi2 = 0.0000

Cameron & Trivedi's decomposition of IM-test

Source	chi2	df	p
Heteroskedasticity	56.04	9	0.0000
Skewness	32.35	3	0.0000
Kurtosis	16.36	1	0.0001
Total	104.76	13	0.0000

Multicollinearity

```
. estat vif
```

Variable	VIF	1/VIF
ln_CreditC~s	9.26	0.107949
ln_CPI	9.10	0.109938
FEDFUNDS	1.29	0.773757
Mean VIF	6.55	

Autocorrelation

```
. estat bgodfrey, lags(1 2 3 4)
```

Breusch-Godfrey LM test for autocorrelation

lags(p)	chi2	df	Prob > chi2
1	276.632	1	0.0000
2	277.313	2	0.0000
3	277.420	3	0.0000
4	277.724	4	0.0000

H0: no serial correlation

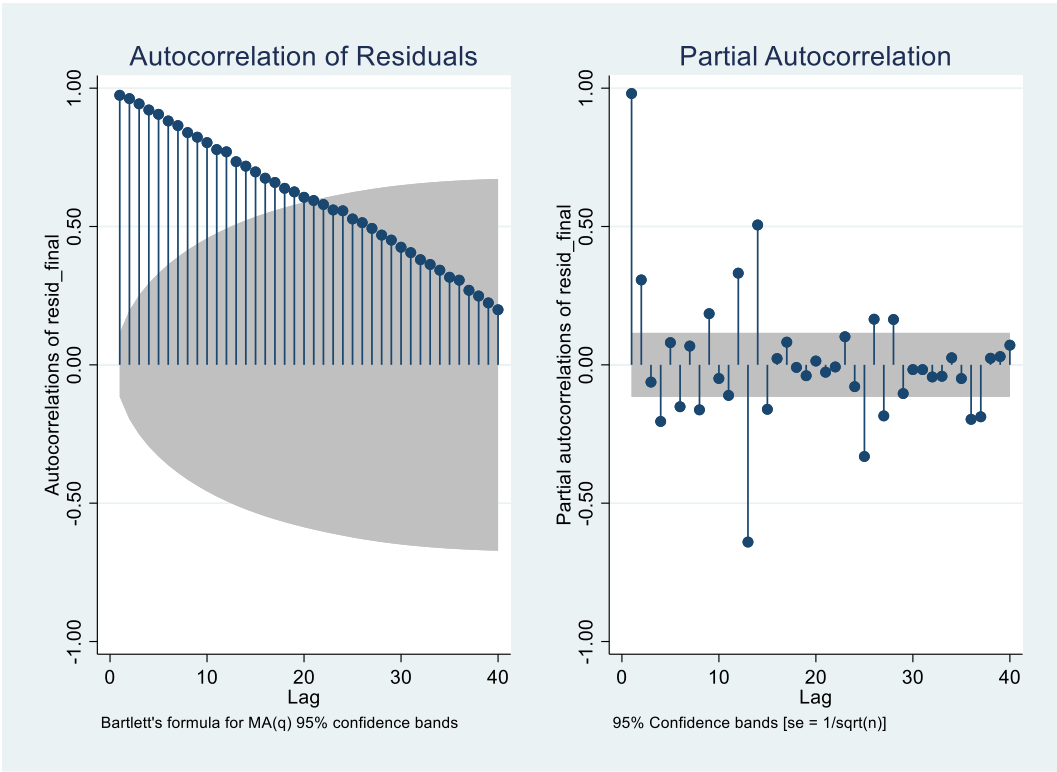
HAC TEST

```
. newey ln_CURRCIR ln_CreditCards ln_CPI FEDFUNDS, lag(12)
```

Regression with Newey-West standard errors Number of obs = 288
Maximum lag = 12 F(3, 284) = 309.00
Prob > F = 0.0000

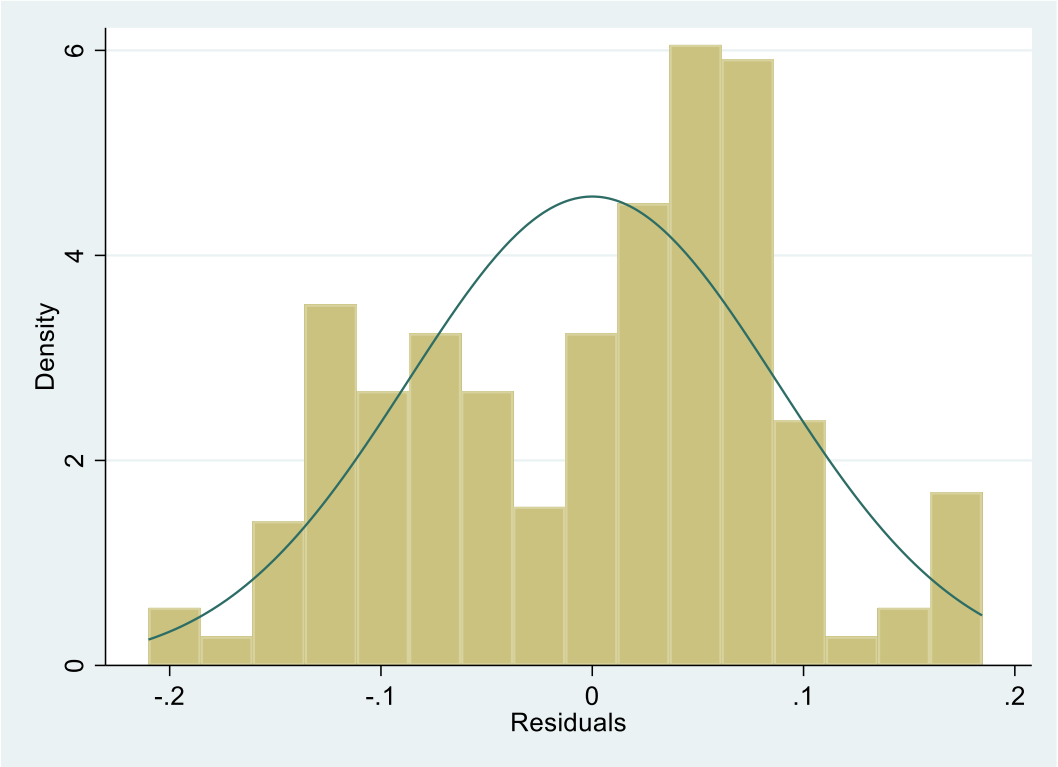
ln_CURRCIR	Newey-West		t	P> t	[95% conf. interval]	
	Coefficient	std. err.				
ln_CreditCards	.0927095	.0977786	0.95	0.344	-.0997533	.2851723
ln_CPI	2.429981	.2996344	8.11	0.000	1.840195	3.019767
FEDFUNDS	-1.574363	.8835522	-1.78	0.076	-3.313505	.1647791
_cons	-13.0854	1.54672	-8.46	0.000	-16.12989	-10.04091

AR/PA



Normality

Shapiro-Wilk W test for normal data					
Variable	Obs	W	V	z	Prob>z
resid_final	288	0.96742	6.694	4.454	0.00000



Stationarity

```
. dfuller ln_CURRCIR, trend
```

Dickey-Fuller test for unit root Number of obs = 287
Variable: ln_CURRCIR Number of lags = 0

H0: Random walk with or without drift

Test statistic	Dickey-Fuller critical value			
	1%	5%	10%	
Z(t)	-0.760	-3.989	-3.429	-3.130

MacKinnon approximate p -value for Z(t) = 0.9689.

```
.  
. dfuller ln_CreditCards, trend
```

Dickey-Fuller test for unit root Number of obs = 287
Variable: ln_CreditCards Number of lags = 0

H0: Random walk with or without drift

Test statistic	Dickey-Fuller critical value			
	1%	5%	10%	
Z(t)	-10.077	-3.989	-3.429	-3.130

MacKinnon approximate p -value for Z(t) = 0.0000.

```
.  
. dfuller ln_CPI, trend
```

Dickey-Fuller test for unit root Number of obs = 287
Variable: ln_CPI Number of lags = 0

H0: Random walk with or without drift

Test statistic	Dickey-Fuller critical value			
	1%	5%	10%	
Z(t)	0.297	-3.989	-3.429	-3.130

MacKinnon approximate p -value for Z(t) = 0.9963.

```
.  
. dfuller FEDFUNDS, trend
```

Dickey-Fuller test for unit root Number of obs = 287
Variable: FEDFUNDS Number of lags = 0

H0: Random walk with or without drift

Test statistic	Dickey-Fuller critical value			
	1%	5%	10%	
Z(t)	-1.554	-3.989	-3.429	-3.130

MacKinnon approximate p -value for Z(t) = 0.8099.

```
.
```

KPSS

```
. kpss ln_CURRCIR, trend
```

KPSS test for ln_CURRCIR

Maxlag = 15 chosen by Schwert criterion
Autocovariances weighted by Bartlett kernel

Critical values for H0: ln_CURRCIR is trend stationary

10%: 0.119 5% : 0.146 2.5%: 0.176 1% : 0.216

Lag order	Test statistic
0	4.48
1	2.25
2	1.51
3	1.14
4	.914
5	.765
6	.66
7	.581
8	.52
9	.471
10	.431
11	.398
12	.37
13	.347
14	.326
15	.308

```
.  
. kpss ln_CreditCards, trend
```

KPSS test for ln_CreditCards

Maxlag = 15 chosen by Schwert criterion
Autocovariances weighted by Bartlett kernel

Critical values for H0: ln_CreditCards is trend stationary

10%: 0.119 5% : 0.146 2.5%: 0.176 1% : 0.216

Lag order	Test statistic
0	1.88
1	1.28
2	.925
3	.724
4	.608
5	.524
6	.466
7	.419
8	.384
9	.354
10	.329
11	.308
12	.288
13	.272
14	.258
15	.245

```
. kpss ln_CPI, trend
```

KPSS test for ln_CPI

Maxlag = 15 chosen by Schwert criterion
Autocovariances weighted by Bartlett kernel

Critical values for H0: ln_CPI is trend stationary

10%: 0.119 5% : 0.146 2.5%: 0.176 1% : 0.216

Lag order	Test statistic
0	2.79
1	1.4
2	.943
3	.712
4	.574
5	.482
6	.416
7	.367
8	.329
9	.298
10	.274
11	.253
12	.236
13	.221
14	.208
15	.197

```
.  
. kpss FEDFUNDS, trend
```

KPSS test for FEDFUNDS

Maxlag = 15 chosen by Schwert criterion
Autocovariances weighted by Bartlett kernel

Critical values for H0: FEDFUNDS is trend stationary

10%: 0.119 5% : 0.146 2.5%: 0.176 1% : 0.216

Lag order	Test statistic
0	3.03
1	1.53
2	1.03
3	.776
4	.627
5	.527
6	.457
7	.404
8	.363
9	.331
10	.305
11	.283
12	.265
13	.25
14	.237
15	.225

```
.
```

Stationarity and Unit Root Tests

. dfuller D.ln_CURRCIR

Dickey-Fuller test for unit root Number of obs = 286
Variable: D.ln_CURRCIR Number of lags = 0

H0: Random walk without drift, d = 0

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-10.837	-3.457	-2.879	-2.570

MacKinnon approximate *p*-value for Z(t) = 0.0000.

.
. dfuller D.ln_CreditCards

Dickey-Fuller test for unit root Number of obs = 286
Variable: D.ln_CreditCards Number of lags = 0

H0: Random walk without drift, d = 0

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-35.158	-3.457	-2.879	-2.570

MacKinnon approximate *p*-value for Z(t) = 0.0000.

.
. dfuller D.ln_CPI

Dickey-Fuller test for unit root Number of obs = 286
Variable: D.ln_CPI Number of lags = 0

H0: Random walk without drift, d = 0

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-9.817	-3.457	-2.879	-2.570

MacKinnon approximate *p*-value for Z(t) = 0.0000.

.
. dfuller D.FEDFUNDS

Dickey-Fuller test for unit root Number of obs = 286
Variable: D.FEDFUNDS Number of lags = 0

H0: Random walk without drift, d = 0

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-7.340	-3.457	-2.879	-2.570

MacKinnon approximate *p*-value for Z(t) = 0.0000.


```
. kpsstest D.ln_CURRCIR
```

KPSS test for D.ln_CURRCIR

Maxlag = 15 chosen by Schwert criterion
Autocovariances weighted by Bartlett kernel

Critical values for H0: D.ln_CURRCIR is trend stationary

10%: 0.119 5% : 0.146 2.5%: 0.176 1% : 0.216

Lag order	Test statistic
0	.415
1	.292
2	.25
3	.223
4	.202
5	.186
6	.172
7	.162
8	.153
9	.147
10	.142
11	.138
12	.132
13	.127
14	.124
15	.121

```
. kpsstest D.ln_CreditCards
```

KPSS test for D.ln_CreditCards

Maxlag = 15 chosen by Schwert criterion
Autocovariances weighted by Bartlett kernel

Critical values for H0: D.ln_CreditCards is trend stationary

10%: 0.119 5% : 0.146 2.5%: 0.176 1% : 0.216

Lag order	Test statistic
0	.00446
1	.0118
2	.0162
3	.0149
4	.023
5	.02
6	.0292
7	.0253
8	.0352
9	.037
10	.0315
11	.0797
12	.0362
13	.0476
14	.0503
15	.0433

```
. kpss D.ln_CPI
```

KPSS test for D.ln_CPI

Maxlag = 15 chosen by Schwert criterion
Autocovariances weighted by Bartlett kernel

Critical values for H0: D.ln_CPI is trend stationary

10%: 0.119 5% : 0.146 2.5%: 0.176 1% : 0.216

Lag order	Test statistic
0	.387
1	.261
2	.227
3	.211
4	.199
5	.19
6	.183
7	.176
8	.17
9	.166
10	.16
11	.155
12	.151
13	.149
14	.147
15	.144

```
.  
. kpss D.FEDFUNDS
```

KPSS test for D.FEDFUNDS

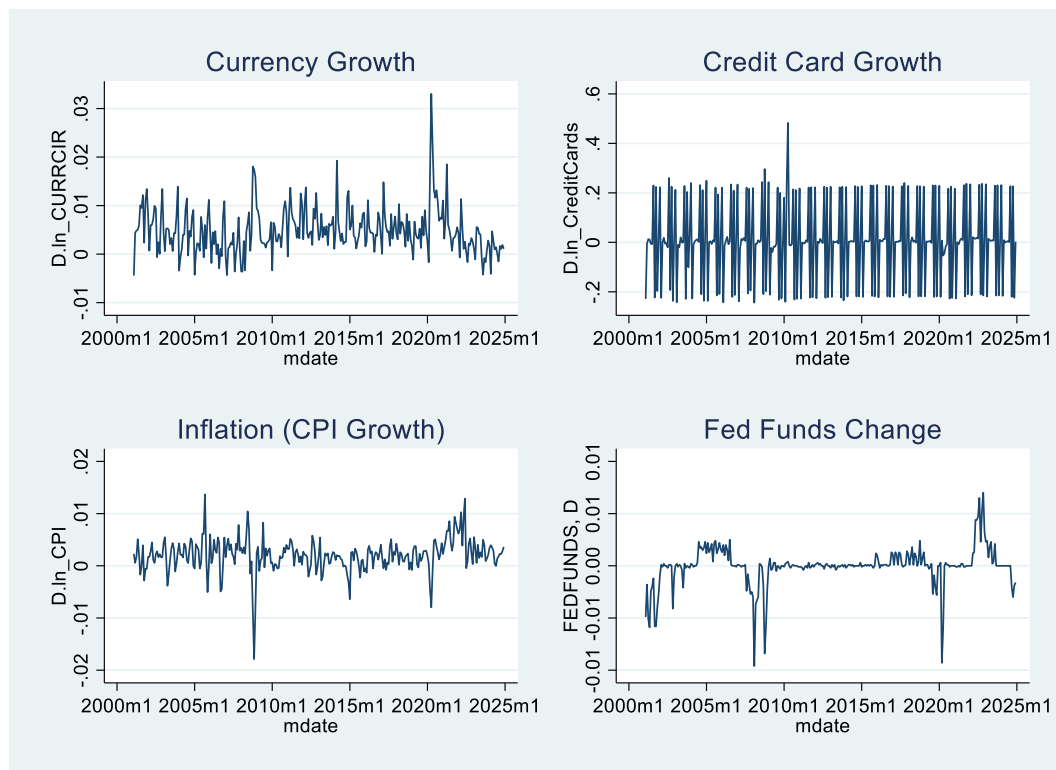
Maxlag = 15 chosen by Schwert criterion
Autocovariances weighted by Bartlett kernel

Critical values for H0: D.FEDFUNDS is trend stationary

10%: 0.119 5% : 0.146 2.5%: 0.176 1% : 0.216

Lag order	Test statistic
0	.293
1	.175
2	.132
3	.108
4	.0929
5	.0821
6	.0738
7	.0674
8	.0623
9	.0582
10	.0549
11	.0523
12	.0501
13	.0483
14	.0468
15	.0455

Stationarity check Graph



Multicollinearity issue resolved

```
. estat vif
```

Variable	VIF	1/VIF
FEDFUNDS		
D1.	1.09	0.914387
ln_CPI		
D1.	1.09	0.918254
ln_CreditC~s		
D1.	1.01	0.994480
Mean VIF	1.06	

Johansen Test

Johansen tests for cointegration

Trend: Constant

Number of obs = 286

Sample: 2001m3 thru 2024m12

Number of lags = 2

Maximum				Trace	Critical
rank	Params	LL	Eigenvalue	statistic	value
0	20	4196.6509	.	65.5927	47.21
1	27	4214.8984	0.11980	29.0977*	29.68
2	32	4226.165	0.07576	6.5644	15.41
3	35	4228.914	0.01904	1.0664	3.76
4	36	4229.4472	0.00372		

Maximum			Eigenvalue		Critical
rank	Params	LL	Maximum		value
0	20	4196.6509	.	36.4951	27.07
1	27	4214.8984	0.11980	22.5333	20.97
2	32	4226.165	0.07576	5.4980	14.07
3	35	4228.914	0.01904	1.0664	3.76
4	36	4229.4472	0.00372		

* selected rank

VECM

Cointegrating equations

Equation	Parms	chi2	P>chi2
_ce1	3	41.91479	0.0000

Identification: beta is exactly identified

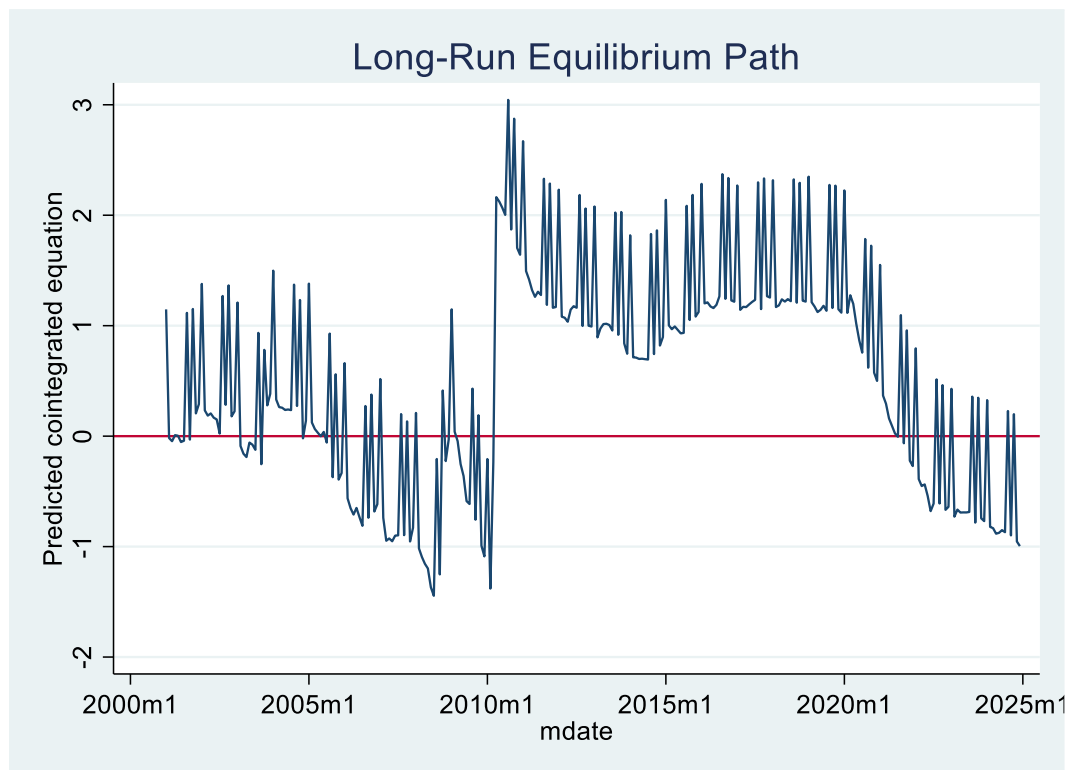
Johansen normalization restriction imposed

beta	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
_ce1						
ln_CURRCIR	1	.	.	.	.	.
ln_CreditCards	4.972568	1.019178	4.88	0.000	2.975015	6.97012
ln_CPI	-17.08382	3.164693	-5.40	0.000	-23.28651	-10.88114
FEDFUNDS	-3.221095	9.63688	-0.33	0.738	-22.10903	15.66684
_cons	89.44094	.	.	.	.	.

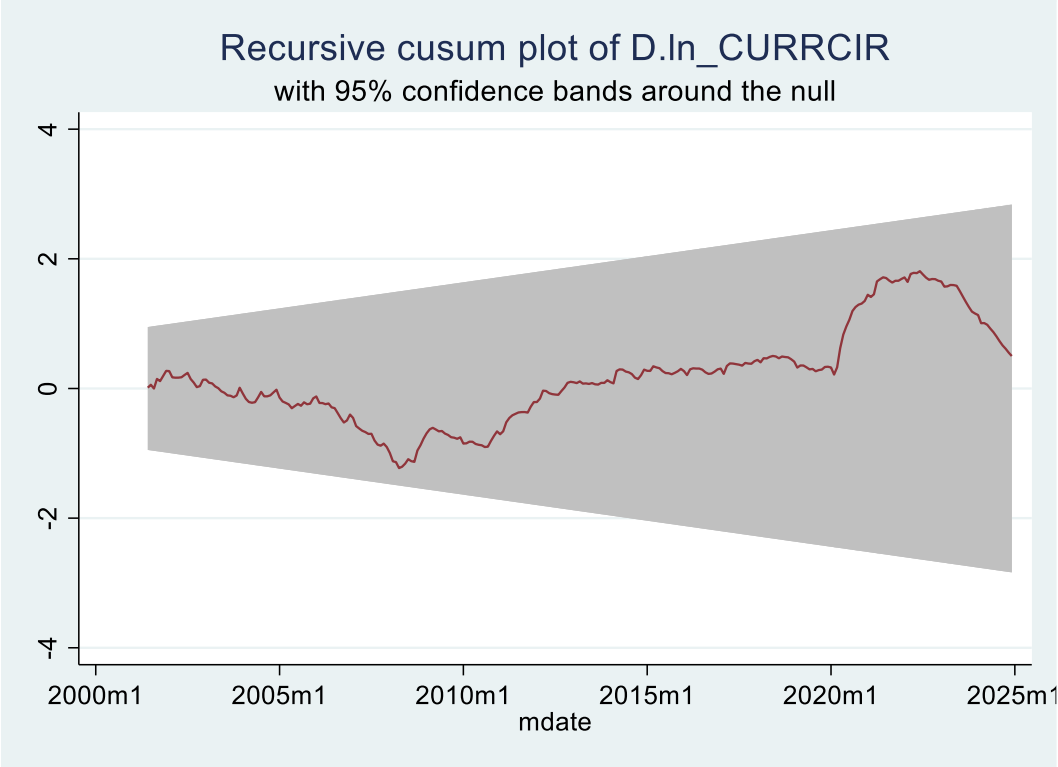
Short-Run Causality Test (Wald Test)

```
test [D_ln_CURRCIR]: LD.ln_CreditCards  
  
( 1) [D_ln_CURRCIR]LD.ln_CreditCards = 0  
  
      chi2( 1) =    12.96  
      Prob > chi2 =    0.0003
```

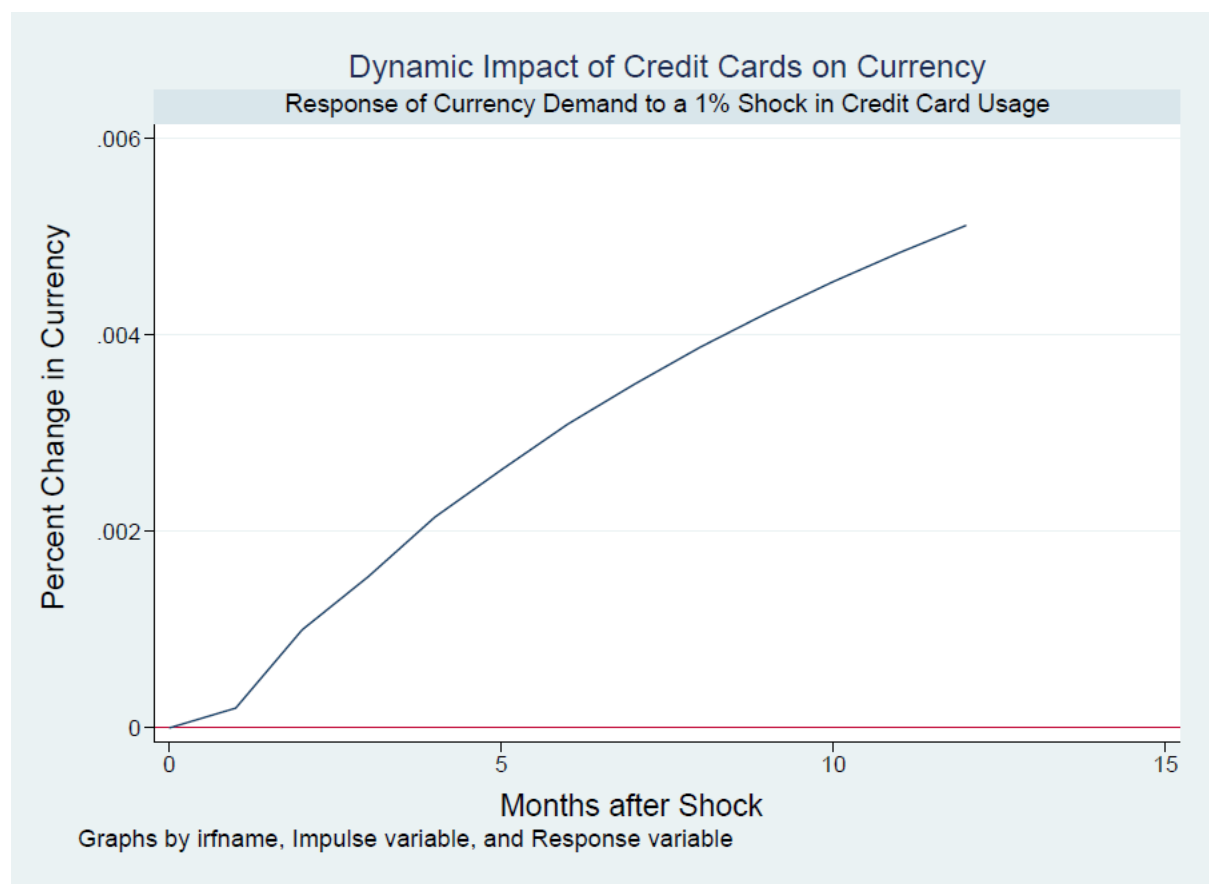
Long Term Graph



CUSUM



IRF



ARIMA

ARIMA regression

Sample: 2001m2 thru 2024m12

Number of obs = 287

Wald chi2(6) = 2144.61

Log likelihood = -1078.325

Prob > chi2 = 0.0000

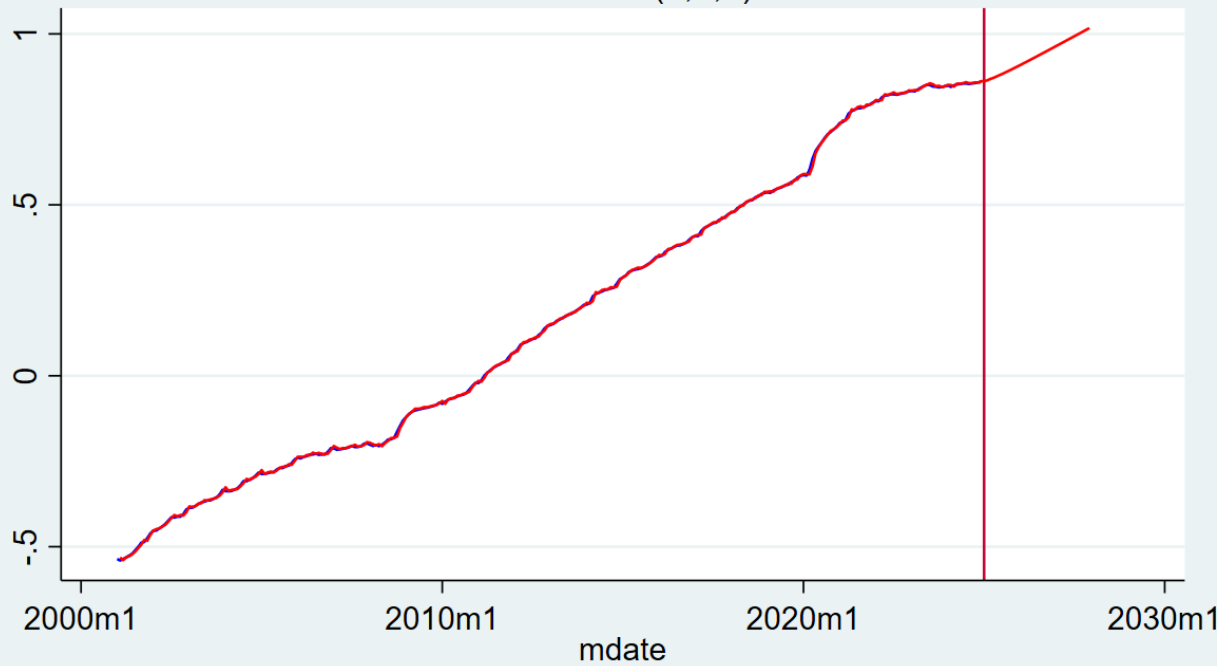
D.CURRCIR		OPG		z	P> z	[95% conf. interval]	
	Coefficient	std. err.					
CURRCIR							
_cons	.0058835	.0020885	2.82	0.005	.0017901	.0099768	
ARMA							
ar							
L1.	-.3453955	.1237861	-2.79	0.005	-.5880118	-.1027793	
L2.	.9089357	.0377542	24.08	0.000	.8349389	.9829326	
L3.	.2892245	.1047501	2.76	0.006	.083918	.494531	
ma							
L1.	.894465	.120009	7.45	0.000	.6592516	1.129678	
L2.	-.6126085	.1066441	-5.74	0.000	-.8216271	-.40359	
L3.	-.5731451	.0564215	-10.16	0.000	-.6837291	-.462561	
/sigma	.0056354	.0001382	40.77	0.000	.0053645	.0059063	

Note: The test of the variance against zero is one sided, and the two-sided confidence interval is truncated at zero.

Forecast Graph

Currency Demand Forecast (2025-2027)

Based on ARIMA(3,1,3) Model



— Actual Data — Forecast